

```
In [1]: from tensorflow.keras.datasets import imdb
```

```
In [2]: (train_data, train_label), (test_data, test_label) = imdb.load_data(num_words = 10000)
```

Out:

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 [=====] - 0s 0us/step
```

```
In [3]: import numpy as np
```

```
def vectorize_sequences(sequences, dimensions = 10000):
    results = np.zeros((len(sequences), dimensions))
    for i, sequences in enumerate(sequences):
        results[i, sequences] = 1
    return results
```

```
x_train = vectorize_sequences(train_data)
y_train = vectorize_sequences(test_data)
```

```
In [4]: y_train = np.asarray(train_label).astype('float32')
y_test = np.asarray(test_label).astype('float32')
```

```
In [5]: from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

```
In [6]: model = Sequential()
model.add(Dense(16, input_shape=(10000, ), activation = "relu"))
model.add(Dense(16, activation = "relu"))
model.add(Dense(1, activation = "sigmoid"))
```

```
In [7]: model.compile(optimizer='adam', loss = 'mse', metrics = ['accuracy'])
```

```
In [8]: model.summary()
```

Out:

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	160016
dense_1 (Dense)	(None, 16)	272
dense_2 (Dense)	(None, 1)	17

```
=====  
Total params: 160,305  
Trainable params: 160,305  
Non-trainable params: 0  
=====
```

```
In [9]: history = model.fit(x_train, y_train, validation_split = 0.3, epochs = 20, verbose = 1, batch_size = 512)
```

Out:

```
Epoch 1/20  
35/35 [=====] - 2s 39ms/step - loss: 0.0041 - accuracy: 0.9960 - val  
_loss: 0.1162 - val_accuracy: 0.8652  
Epoch 2/20  
35/35 [=====] - 1s 35ms/step - loss: 0.0042 - accuracy: 0.9959 - val  
_loss: 0.1171 - val_accuracy: 0.8648  
Epoch 3/20  
35/35 [=====] - 1s 32ms/step - loss: 0.0043 - accuracy: 0.9958 - val  
_loss: 0.1192 - val_accuracy: 0.8636  
Epoch 4/20  
35/35 [=====] - 1s 31ms/step - loss: 0.0042 - accuracy: 0.9959 - val  
_loss: 0.1210 - val_accuracy: 0.8619  
Epoch 5/20  
35/35 [=====] - 1s 31ms/step - loss: 0.0041 - accuracy: 0.9960 - val  
_loss: 0.1208 - val_accuracy: 0.8628  
Epoch 6/20  
35/35 [=====] - 1s 42ms/step - loss: 0.0041 - accuracy: 0.9959 - val  
_loss: 0.1210 - val_accuracy: 0.8619
```

```
Epoch 7/20
35/35 [=====] - 2s 47ms/step - loss: 0.0040 - accuracy: 0.9961 - val
_loss: 0.1213 - val_accuracy: 0.8621
Epoch 8/20
35/35 [=====] - 1s 31ms/step - loss: 0.0041 - accuracy: 0.9960 - val
_loss: 0.1214 - val_accuracy: 0.8620
Epoch 9/20
35/35 [=====] - 1s 32ms/step - loss: 0.0041 - accuracy: 0.9961 - val
_loss: 0.1204 - val_accuracy: 0.8655
Epoch 10/20
35/35 [=====] - 1s 32ms/step - loss: 0.0041 - accuracy: 0.9961 - val
_loss: 0.1215 - val_accuracy: 0.8633
Epoch 11/20
35/35 [=====] - 1s 37ms/step - loss: 0.0040 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8623
Epoch 12/20
35/35 [=====] - 1s 32ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1223 - val_accuracy: 0.8617
Epoch 13/20
35/35 [=====] - 1s 35ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1223 - val_accuracy: 0.8613
Epoch 14/20
35/35 [=====] - 1s 36ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1223 - val_accuracy: 0.8616
Epoch 15/20
35/35 [=====] - 1s 31ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1223 - val_accuracy: 0.8613
Epoch 16/20
35/35 [=====] - 1s 38ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8611
Epoch 17/20
35/35 [=====] - 2s 48ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8607
Epoch 18/20
35/35 [=====] - 1s 37ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8608
Epoch 19/20
35/35 [=====] - 1s 35ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8605
Epoch 20/20
35/35 [=====] - 2s 60ms/step - loss: 0.0039 - accuracy: 0.9961 - val
_loss: 0.1224 - val_accuracy: 0.8609
```